

ROS2 ATOSFleetManagement Cheatsheet

Updated: 2026-04-23

1) Build

Option A (recommended helper script)

```
cd <ATOS_REPO>
./scripts/build_atosfleetmanagement.sh
```

Option B (explicit colcon)

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```

2) Run ATOSFleetManagement

Without simulators

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True with_truck_simulator:=False
```

With simulators

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True with_truck_simulator:=True
```

GUI: <http://localhost:8420> (hard refresh with Ctrl+Shift+R after UI updates).

3) Runtime control for simulators

Stop SIM-2 while launch keeps running

```
pkill -f "__node:=atos_truck_simulator_2"
```

Restart SIM-2 manually

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 run atos atos_truck_simulator --ros-args \
  -r __ns:=/atos \
  -r __node:=atos_truck_simulator_2 \
  -p uid:=L5S-TRUCK-SIM-2 \
  -p start_index:=250 \
  -p target_speed_kmh:=40.0 \
  -p acceleration_mps2:=2.0
```

4) Testing and diagnostics

Inspect state once:

```
ros2 topic echo /atos/truck_objects/state --once
```

Important fields: last_cot_message, last_tcp_command, last_tcp_warning, tcp_connected, path_name, path_index, speed_mps.

Typical warning texts: TCP client disconnected; No active TCP socket for this truck; TCP send failed (errno=...); TLS send failed (ssl_error=...).

5) Speed units

Simulators send CoT speed in km/h. TruckObjectControl uses speed_mps internally. GUI displays km/h.

6) Quick rebuild after changes

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```

7) Example local paths

: /home/sepast/Documents/repos/ATOS

: /home/sepast/atos_ws